The Art and Science of Transportation Research in the AI Era

Welcome to ASTRAI

Meng Cai



A stray
/ə streɪ/

Ashtray
/ˈæʃ treɪ/

Learning goals

#1 Locate course information and materials

#2 Explain what ASTRAI is about

#3 Explain what you need to do in ASTRAI

Agenda

#1 Getting to know each other

#2 What is ASTRAI

#3 What you need to do

#4 What ASTRAI can lead to

Agenda

TECHNISCHE UNIVERSITÄT DARMSTADT

IVV Institut für Verkehrsplanung und Verkehrstechnik TU Darmstadt

#1 Getting to know each other

#2 What is ASTRAI

#3 What you need to do

#4 What ASTRAI can lead to

#1.1 Name tent



Let’s make a name tent first.

#1.2 About you

Please share:

- Your name, and the meaning or story behind it
- Your background, e.g., your major and semester
- How you found out about this course
- What you hope to get out of it

#1.3 About me

Name: Meng Cai 蔡 蒙

Background: Ph.D. in Urban Planning (2024), Michigan State University, specializing in computational modelling

Research interests:

1. Urban sustainability and technologies
2. Influence of AI adoption in urban systems
3. Human behavior and wellbeing through virtual reality



#1.2 About you

Please share:

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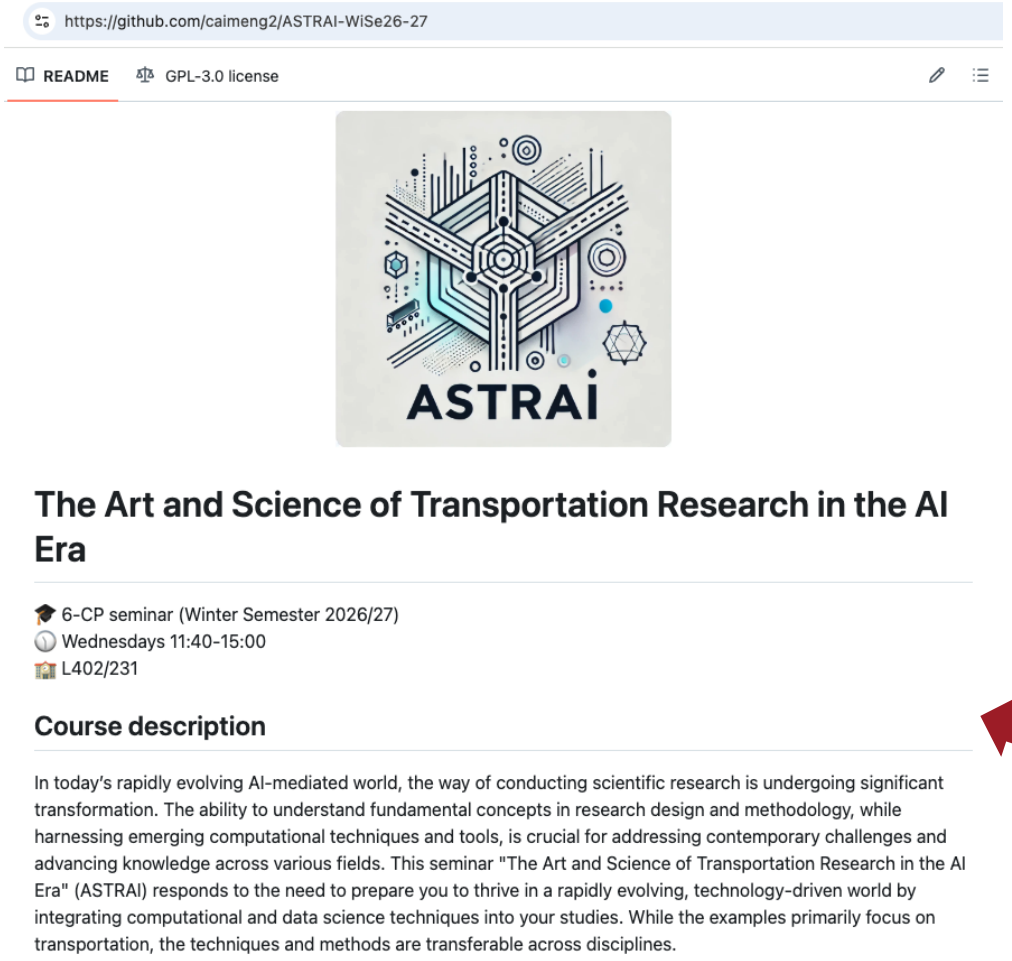
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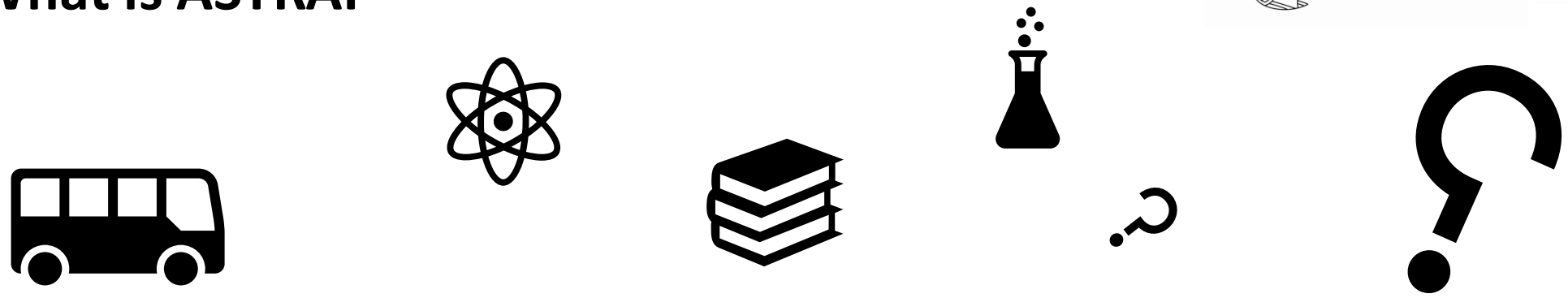
#2.1 Basic info



What: 6-CP seminar
When: Wednesdays 11:40-15:00
Where: L402/231
Contact: cai@verkehr.tu-darmstadt.de
Course website:
https://github.com/caimeng2/ASTRAI-WiSe26-27
Everything in the repo is freely and publicly available under the GPL-3.0 license.
No Moodle page.
I will send you messages through TUCaN.

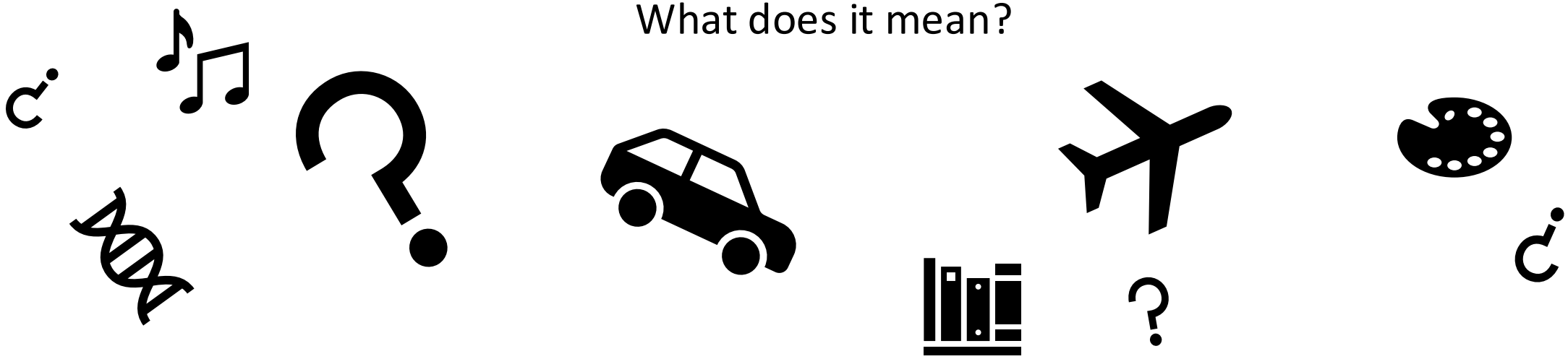
Course contents in GitHub repo!

#2.2 What is ASTRAI



The Art and Science of Transportation Research in the AI Era

What does it mean?

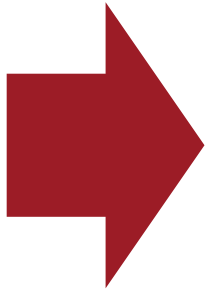


#2.2 What is ASTRAI

The Art and Science of Transportation **Research in the AI Era**

Research processes:

- Finding a research topic
- Defining a research question
- Conducting a literature review
- Designing a methodology
- Collecting data
- Analyzing data
- Interpreting results
- Report preparation and presentation of findings



- Interdisciplinary topics
- Data-driven
- AI-powered literature review
- New methods and tools
- New data sources, automated data collection, real-time data
- Easier and better visualization
- AI-assisted writing

#2.2 What is ASTRAI

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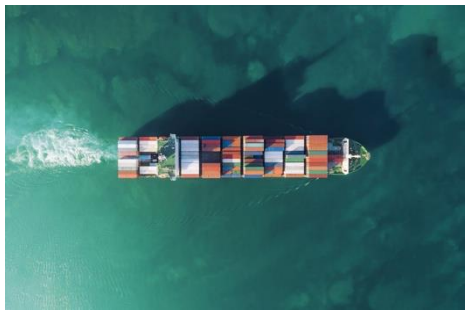
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The Art and Science of **Transportation** Research in the AI Era

The examples are mostly transportation-related, but the techniques and methods are transferable across disciplines.





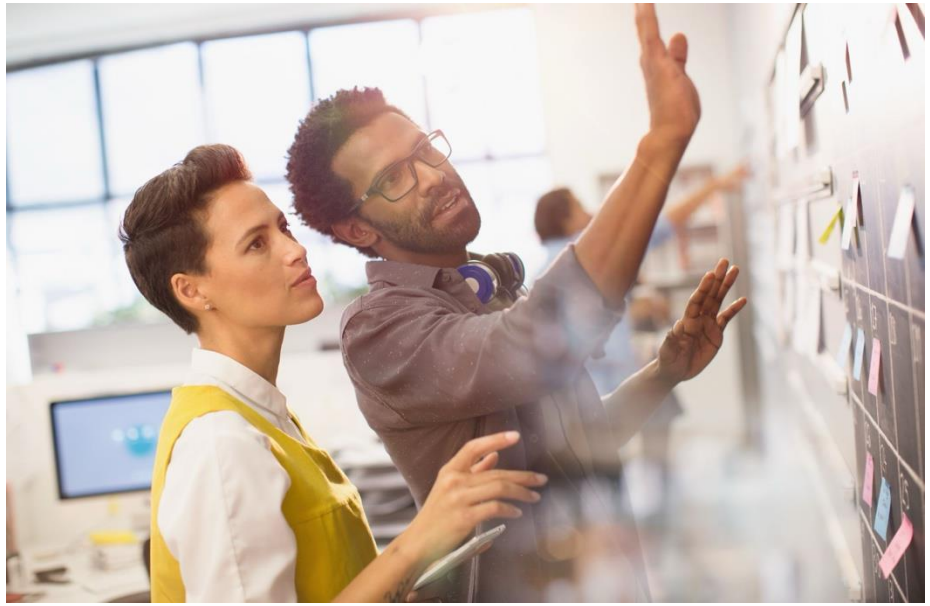




Picture sources: Microsoft stock images.

#2.2 What is ASTRAI

The **Art** and Science of Transportation Research in the AI Era







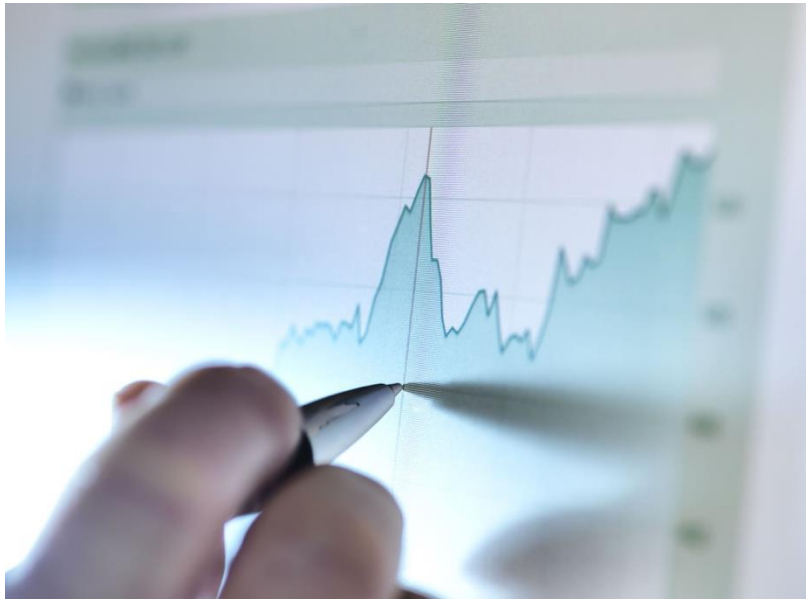
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"Art" refers to the **creative** aspects of research

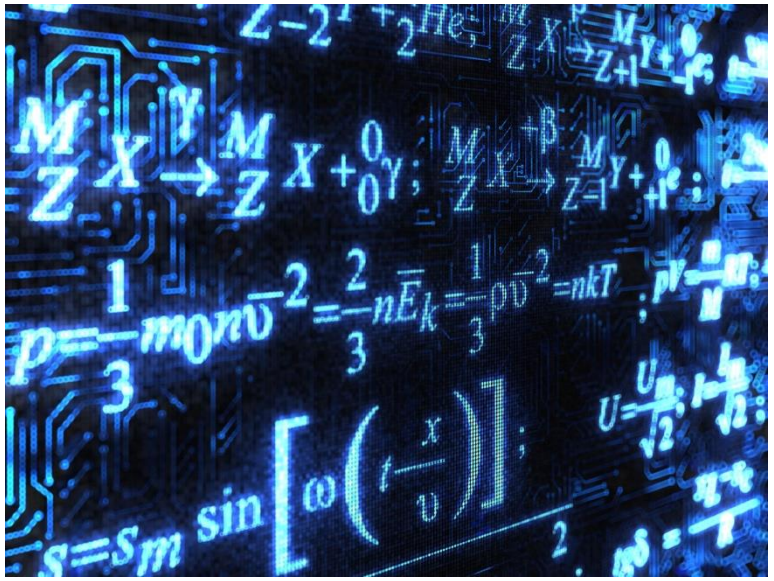
- Asking the right questions
- Developing novel approaches
- Finding innovative solutions
- Effectively communicating through compelling visuals
- Drawing inspiration from guest speakers

#2.2 What is ASTRAI

The Art and **Science** of Transportation Research in the AI Era







Picture sources: Microsoft stock images.

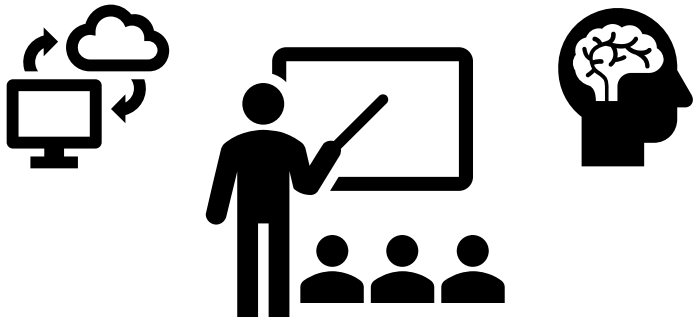
"Science" refers to the **tools, techniques, and knowledge foundation** for research

- Developing a good research design
- Having robust methodologies
- Leveraging advanced computational tools
- Be able to tell if the answers provided by AI are correct

#2.3 ASTRAI vision

To develop the right **mindset**:

- work across disciplines instead of domain-specific fields
- Stay updated
- Maintain a healthy skepticism of AI-generated results, recognizing limitations
- Address ethical issues proactively in research design and implementation
- Open science



To develop new **skillsets**:

- Programming proficiency
  - Understand and code in commonly used languages such as Python
  - Get familiar with version control
- Data handling
  - Practice techniques for data cleaning, processing, and analyzing
- Data visualization
  - Present data insights effectively via visualization tools

#2.4 Tentative schedule

- Week 1 (Oct 14, 2026): Welcome
- Week 2 (Oct 21, 2026): Git; presenting your initial ideas for YOU-lead
- Week 3 (Oct 28, 2026): Traffic flow simulation (guest speaker)
- Week 4 (Nov 4, 2026): Gaming for science (guest speaker)
- Week 5 (Nov 11, 2026): Python basics (self-study session) or R basics?
- Week 6 (Nov 18, 2026): Web scraping
- Week 7 (Nov 25, 2026): Data wrangling; signing up for a YOU-lead slot
- Week 8 (Dec 2, 2026): Open-source tools
- Week 9 (Dec 9, 2026): Research design and methodology; confirming your topic for YOU-lead
- Week 10 (Dec 16, 2026): Machine learning, AI, and LLMs
- 2027: Your turn

#2.5 What ASTRAI is NOT

- It is not a lecture series.
- It is not a programming course.
- It is not aimed at making you a programmer, a computer scientist, or a computational engineer

Instead, ASTRAI is:

- **Hands-on**: you cannot just sit and listen
- **Exploratory**: we explore methods and AI together
- **Open**: questions, ideas, and disagreement are all welcome. Feel free to interrupt me anytime if something is not clear



Picture sources: Microsoft stock images.

#2.6 Learning outcomes

By the end of the course, you will:

- Illustrate the fundamental concepts in research design and methodology
- Effectively formulate research questions and design a study
- Develop proficiency in data science tools and techniques



Picture sources: Microsoft stock images.

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#3.1 What you need to do

Participation	• Attend class every week • Bring your laptop and power cord
Pre-class requirement	From time to time, install software and/or setup accounts
In-class activities	Practice coding, debugging, discussion, etc.
Student-led learning (YOU-lead)	You learn and teach the method and software behind your proposal (i.e., your exam)
Final examination	Develop a one-page research proposal incorporating computational methods

#3.2 Final examination

A written **research proposal** on a topic of your choice

- The methodology must use computational tools and techniques
- Maximum 1 page, excluding references and the AI use acknowledgment
- Deadline: February 19, 2027

- Next session (Oct 21): presenting your initial ideas



#3.2 Final examination

Recommended structure

- **Introduction**: the problem, why it matters, and the research gap
- **Research questions**: at most three, with hypotheses if applicable
- **Methodology**: data sources, computational tools, and each step of your approach
- **Limitations**: biases, data gaps, assumptions

- References: APA 7th edition, with DOIs or URLs
- AI use acknowledgment



#3.2 Final examination

AI Use Acknowledgment

- **Which AI tools** you used
- **What** you used them for (e.g., learning a method, searching the literature, language editing)
- **Where** they affected the proposal, and **how** you checked the output (e.g., introduction: verifying every reference against the original source)

- If you did not use AI at all, say so.

- **Undeclared AI use or fully AI-generated text counts as cheating and results in a fail (5.0).**



#3.2 Final examination

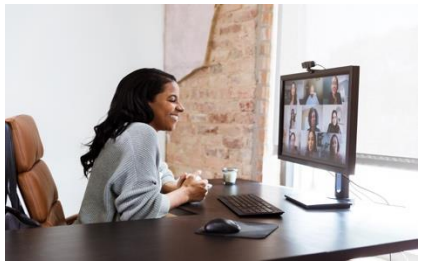
Grading

- Introduction: 20%
- Research questions: 20%
- Methodology: 20%
- Limitations: 20%
- Writing (incl. references and AI acknowledgment): 20%



#3.3 YOU-lead

- You choose one computational method for your research proposal (e.g., traffic simulation)
- You study the method, and the software used to implement it (e.g., SUMO)
- You teach the method and the software to your classmates in class





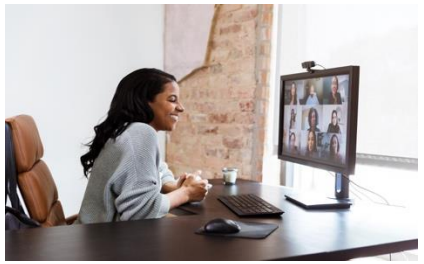




#3.3 YOU-lead

Why?

- **Proven method**: learning by teaching is well established. Explaining something to others deepens your own understanding.
- **Collaboration**: we learn from each other, not just from me
- **Exam**: your presentation becomes the foundation of your proposal
- **Bonus**: a good presentation can improve your final grade by up to 0.4









#3.4 Questions?

Do you remember where to find all the course information and materials?

Take a moment to review the repo. Do you have any questions?

#3.5 Before next class: Install Git

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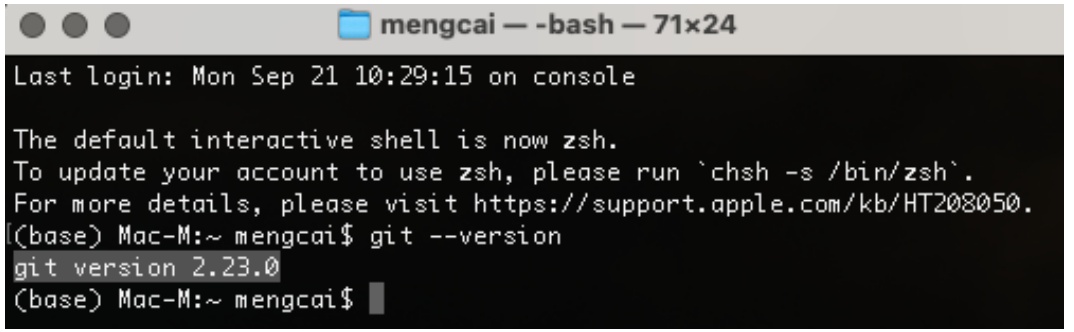
- **Instructions**: https://github.com/git-guides/install-git



- **Check if Git is installed**: open Terminal (Mac/Linux) or Git Bash (Windows) and type:

  git --version

  You should see a version number
  The number itself does not matter



We'll spend the first 10 minutes of next week's class helping each other with installation problems, if any.

#3.5 Before next class: Your idea



Come up with an initial idea for your final proposal/ YOU-lead:

- What problem would you like to solve?
- What research question would you like to answer?
- **Which computational method would you like to learn and teach?**

Present your idea next week: about 10 minutes per person, including questions, discussion, and feedback.

Your idea doesn't need to be polished. It can, and probably will, change over the semester.

#3.6 Questions?

Any questions before we move on to the last part?

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#4 Where ASTRAI can take you

A former ASTRAI student from last semester will share his experience.

Let’s go to the **scAInce Lab** (L101/K163).

